ModelIt!

Student Activity Pack

The Antibiotic Resistance Arms Race

How Superbugs Are Winning — And Why We Created Them

Grade 09L2 | NGSS HS-LS4-2, HS-LS4-3 | Cause and Effect: Mechanism and Prediction

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What are antibiotics and what do they do? Have you ever taken them?

2 Why do you think doctors always say to finish your entire prescription even if you feel better?

3 What is a 'superbug' and why might it be dangerous?

4 Draw or describe what you think happens inside your body when you take an antibiotic.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Antibiotic Dosage | Treatment Duration | Bacterial Population | Resistant Bacteria Percentage | Mutation Rate | Immune Response

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Full Course

High dosage, full duration, normal immune response

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Incomplete Course

High dosage, 50% duration, normal immune response

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Sub-Therapeutic Dosage

Low dosage, full duration, normal immune response

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Horizontal Gene Transfer Rate — The rate at which bacteria share resistance genes with other bacteria through plasmid transfer, even between different species

Antibiotic Diversity — The number of different antibiotic classes available for treatment — when resistance develops to one, doctors switch to another

Biofilm Formation — The ability of bacteria to form protective communities on surfaces, which dramatically increases their resistance to antibiotics

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why are doctors terrified of "superbugs" — and how did WE create them? Use evidence from your simulation to support your answers.

1. **Why is the reinforcing feedback loop in antibiotic resistance so dangerous compared to a system with balancing feedback?**

2. **Your model shows that incomplete treatment is worse than no treatment. How would you explain this counterintuitive result to a friend?**

3. **How does horizontal gene transfer make the resistance problem even worse than your basic model shows?**

4. **What role does the immune system play in your model, and why can't we rely on antibiotics alone?**

5. **If you were a hospital administrator, what policies would you implement based on your model's predictions?**

8 STEM Challenge: Design a Public Health Antibiotic Stewardship Campaign

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YOUR ENGINEERING MISSION

Using data from your simulation model, design an evidence-based public health campaign that communicates WHY antibiotic misuse creates superbugs and recommends specific behavioral changes to slow resistance evolution.

THE SCENARIO

The World Health Organization has hired your team to create a campaign targeting teenagers and young adults. You must use your model data to create compelling, scientifically accurate messaging about antibiotic resistance that changes behavior, not just awareness.

Engineering Design Thinking

1. WHAT DATA VISUALIZATIONS FROM YOUR MODEL WOULD BE MOST PERSUASIVE IN YOUR CAMPAIGN?
2. HOW WILL YOU COMMUNICATE THE URGENCY WITHOUT CAUSING PANIC OR HOPELESSNESS?
3. WHAT MEDIUM (SOCIAL MEDIA, POSTER, VIDEO, APP) BEST REACHES YOUR TARGET AUDIENCE?
4. HOW WILL YOU MEASURE WHETHER YOUR CAMPAIGN ACTUALLY CHANGES BEHAVIOR, NOT JUST AWARENESS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Epidemiologists and Antimicrobial Resistance Researchers study how diseases spread and how bacteria evolve resistance. They work at the CDC, WHO, pharmaceutical companies, and universities, earning $75,000–$150,000/year. This field is considered one of the most critical public health challenges of the 21st century.